SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I M.Varshitha Reddy ,192525443 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Title:Design and Implementation of an AI-Based Student Support Assistant using Google Dialogflow

1. Aim

To design and implement an AI-based Student Support Assistant using Google Dialogflow that provides instant responses to student queries related to fee payment, scholarships, examination results, faculty contact, and placement services.

2. Problem Analysis and Agent Design

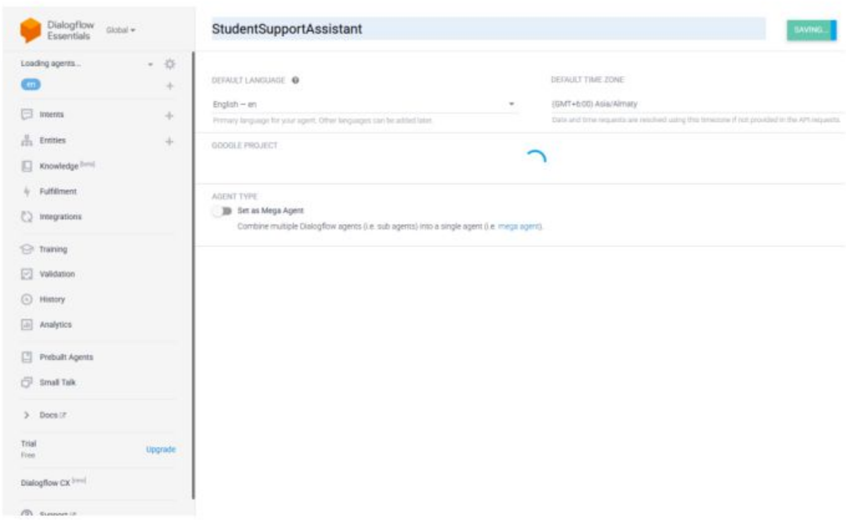
The university receives a large number of student queries every day regarding fee payment, scholarships, examination results, faculty contact, and placement services. Responding to these queries manually takes time and increases the workload of university staff. Students may also experience delays in getting accurate information, especially outside office hours. Therefore, an AI-based Student Support Assistant is required to provide instant, accurate, and 24/7 responses to student queries.

Why Artificial Intelligence?

Artificial Intelligence enables chatbots to understand natural language, identify user intentions, and provide relevant responses automatically. An AI-based chatbot can answer multiple student queries simultaneously, reduce the workload of university staff, improve response speed, and provide consistent information at any time. By using Dialogflow, the chatbot recognizes user intents and entities, making interactions more intelligent and user-friendly.

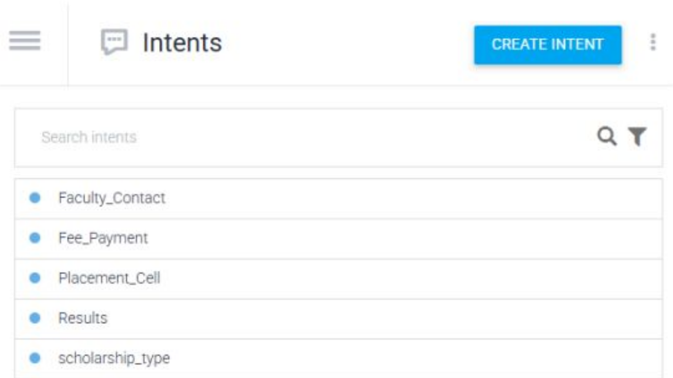
3. Dialogflow Agent Development

Step 1:Created Dialogflow Agent



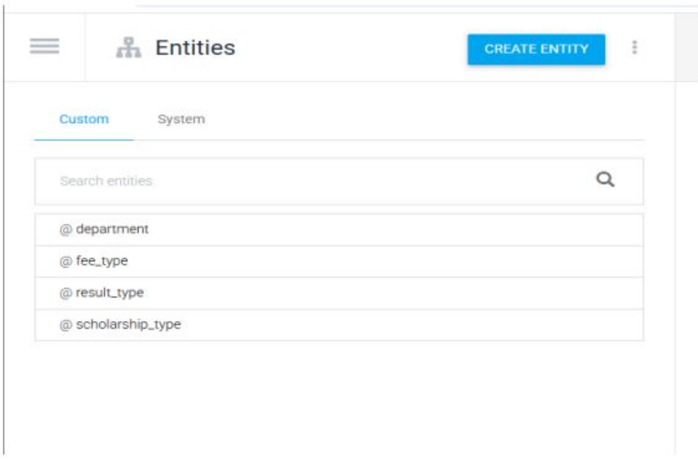
Step 2:Created Intents

- Fee_Payment
- Scholarship_type
- Results
- Faculty_Contact
- Placement_Cell



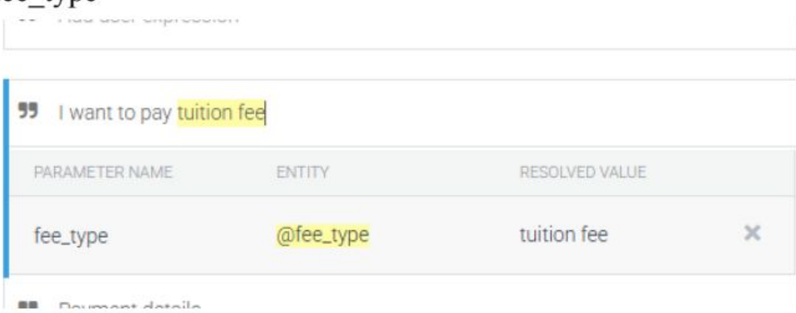
Step 3:Created Entities

Entity	**Values**
fee_type	Tuition, Hostel, Examination, Transport
scholarship_type	Merit, Sports, Minority
result_type	Semester, Mid, Final
department	CSE, AI, ECE, EEE



Step 4:Linked Entities to Intents

Example:
Training Phrase : I want to pay tuition fee
Highlighted: tuition fee
Entity: @fee_type

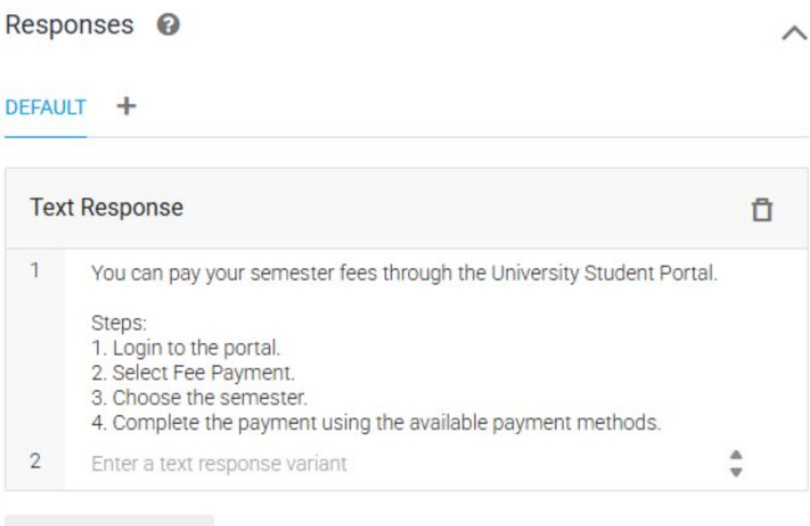


Step 5:Responses

Add reponses
Choose one intent (for example, **Fee_Payment**) and show its response.
Example Response (Fee_Payment):

You can pay your semester fees through the University Student Portal.
Steps:

1. Login to the Student Portal.
2. Select **Fee Payment**.
3. Choose the semester.
4. Complete the payment before the deadline.

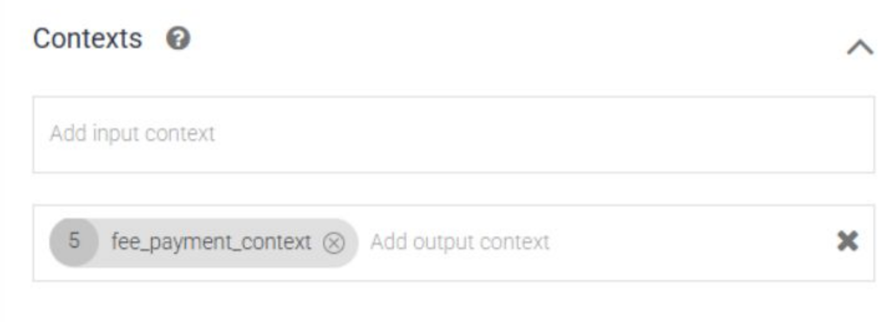


Step 6: Contexts

Intent	**Output Context**
Fee_Payment	fee_payment_context
Scholarship	scholarship_context
Results	result_context
Faculty_Contact	faculty_context
Placement_Cell	placement_context

Example 1: Fee_Payment Intent

Output Context : fee_payment_context
Lifespan**:** 5



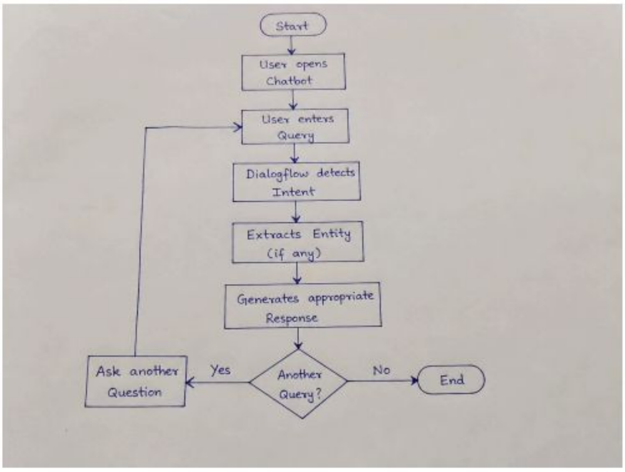
4. Problem Formulation and Conversation Flow

Initial State : User opens chatbot.

Goal State : Student receives the required information.

Actions

- Ask fee details
- Ask scholarship
- Ask result
- Ask faculty contact
- Ask placement details



5. Search Strategy Application

Breadth First Search (BFS)

- User enters query.
- Dialogflow checks all intents.
- Finds the closest matching intent.
- Generates the correct response.

Why BFS?

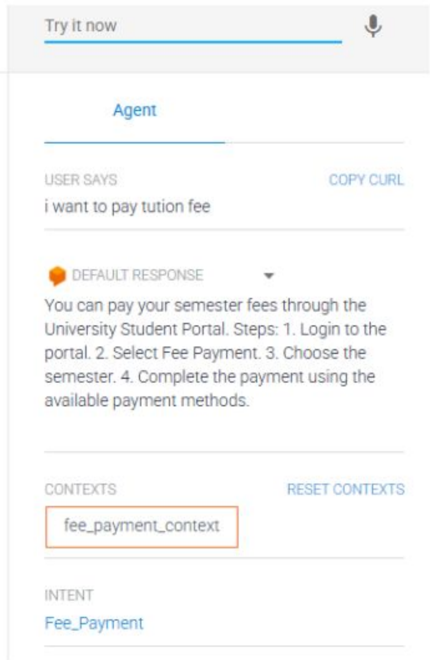
- Easy navigation.
- Fast intent selection.
- Suitable for small chatbot.

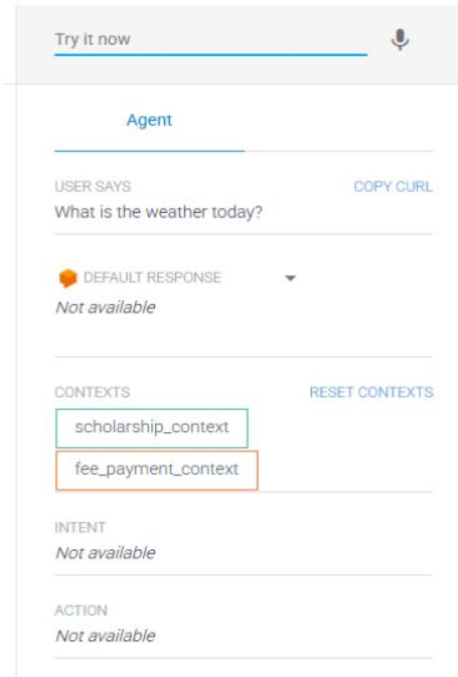
6. Testing and Evaluation

Create a table.

Test Case	User Input	Expected Output	Result
1	Hi	Welcome message	Pass
2	Fee payment	Fee response	Pass
3		Default fallback	Pass

Test Case	User Input	Expected Output	Result
	Weather \today		





7. Conclusion

The AI-based Student Support Assistant was successfully implemented using Google Dialogflow. The chatbot correctly identified user intents, extracted entities, and provided accurate responses to student queries. The chatbot can be further enhanced by integrating it with the university database for real-time information.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.